Foundations of Semi-supervised learning

[Semi-supervised learning]

1.0 Content Block

The data lie approximately on a manifold of much lower dimension than the input space. In this case learning the manifold using both the labeled and unlabeled data can avoid the curse of dimensionality. Then learning can proceed using distances and densities defined on the manifold. The manifold assumption is practical when high-dimensional data are generated by some process that may be hard to model directly, but which has only a few degrees of freedom. For instance, human voice is controlled by a few vocal folds, and images of various facial expressions are controlled by a few muscles. In these cases, it is better to consider distances and smoothness in the natural space of the generating problem, rather than in the space of all possible acoustic waves or images, respectively.

2.0 Content Block

Heuristic approaches Some methods for semi-supervised learning are not intrinsically geared to learning from both unlabeled and labeled data, but instead make use of unlabeled data within a supervised learning framework. For instance, the labeled and unlabeled examples x 1 , … , x l + u {\displaystyle x_{1},\dots ,x_{l+u}} may inform a choice of representation, distance metric, or kernel for the data in an unsupervised first step. Then supervised learning proceeds from only the labeled examples. In this vein, some methods learn a low-dimensional representation using the supervised data and then apply either low-density separation or graph-based methods to the learned representation. Iteratively refining the representation and then performing semi-supervised learning on said representation may further improve performance. Self-training is a wrapper method for semi-supervised learning. First a supervised learning algorithm is trained based on the labeled data only. This classifier is then applied to the unlabeled data to generate more labeled examples as input for the supervised learning algorithm. Generally only the labels the classifier is most confident in are added at each step. In natural language processing, a common self-training algorithm is the Yarowsky algorithm for problems like word sense disambiguation, accent restoration, and spelling correction. Co-training is an extension of self-training in which multiple classifiers are trained on different (ideally disjoint) sets of features and generate labeled examples for one another.

3.0 Content Block

An exact solution is intractable due to the non-convex term (1 − | f (x) |) + {\displaystyle (1-|f(x)|)_{+}} , so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).

4.0 Content Block

f ∗ = argmin f (∑ i = 1 l (1 − y i f (x i)) + + λ 1 ■ h ■ H 2 + λ 2 ∑ i = l + 1 l + u (1 − | f (x i) |) +) {\displaystyle f^{*}={\underset {f}{\operatorname {argmin} }}\left(\displaystyle \sum _{i=1}^{l}(1-y_{i}f(x_{i}))_{+}+\lambda _{1}\|h\|_{\mathcal {H}}^{2}+\lambda _{2}\sum _{i=l+1}^{l+u}(1-|f(x_{i})|)_{+}\right)}

5.0 Content Block

Generative models Generative approaches to statistical learning first seek to estimate p (x | y) {\displaystyle p(x|y)} , the distribution of data points belonging to each class. The probability p (y | x) {\displaystyle p(y|x)} that a given point x {\displaystyle x} has label y {\displaystyle y} is then proportional to p (x | y) p (y) {\displaystyle p(x|y)p(y)} by Bayes' rule. Semi-supervised learning with generative models can be viewed either as an extension of supervised learning (classification plus information about p (x) {\displaystyle p(x)}) or as an extension of unsupervised learning (clustering plus some labels). Generative models assume that the distributions take some particular form p (x | y , θ) {\displaystyle p(x|y,\theta)} parameterized by the vector θ {\displaystyle \theta } . If these assumptions are incorrect, the unlabeled data may actually decrease the accuracy of the solution relative to what would have been obtained from labeled data alone. However, if the assumptions are correct, then the unlabeled data necessarily improves performance. The unlabeled data are distributed according to a mixture of individual-class distributions. In order to learn the mixture distribution from the unlabeled data, it must be identifiable, that is, different parameters must yield different summed distributions. Gaussian mixture distributions are identifiable and commonly used for generative models. The parameterized joint distribution can be written as p (x , y | θ) = p (y | θ) p (x | y , θ) {\displaystyle p(x,y|\theta)=p(y|\theta)p(x|y,\theta)} by using the chain rule. Each parameter vector θ {\displaystyle \theta } is associated with a decision function f θ (x) = argmax y p (y | x , θ) {\displaystyle f_{\theta }(x)={\underset {y}{\operatorname {argmax} }}\ p(y|x,\theta)} . The parameter is then chosen based on fit to both the labeled and unlabeled data, weighted by λ {\displaystyle \lambda } :

6.0 Content Block

Weak Supervision in Predictive Maintenance Weak supervision is an emerging machine learning approach in predictive maintenance that addresses the challenge of limited or imprecise labeled data. Traditional predictive maintenance models often rely on large volumes of accurately labeled failure and operational data, which can be costly or impractical to obtain. Weak supervision mitigates this by leveraging imperfect sources of supervision—such as noisy labels, heuristics, domain expert rules, or partially labeled datasets—to train predictive models. By incorporating techniques such as data programming, label modeling, and semi-supervised learning, weak supervision enables the development of robust predictive maintenance systems capable of identifying equipment failures or anomalies with reduced reliance on high-quality labeled data. This approach is particularly valuable in industrial settings where machine failures are rare and labeled fault data is scarce .

7.0 Content Block

Laplacian regularization Laplacian regularization has been historically approached through graph-Laplacian. Graph-based methods for semi-supervised learning use a graph representation of the data, with a node for each labeled and unlabeled example. The graph may be constructed using domain knowledge or similarity of examples; two common methods are to connect each data point to its k {\displaystyle k} nearest neighbors or to examples within some distance ■ {\displaystyle \epsilon } . The weight W i j {\displaystyle W_{ij}} of an edge between x i {\displaystyle x_{i}} and x j {\displaystyle x_{j}} is then set to e – ■ x i – x j ■ 2 / ■ 2 {\displaystyle e^{-\|x_{i}-x_{j}\|^{2}/\epsilon ^{2}}} . Within the framework of manifold regularization, the graph serves as a proxy for the manifold. A term is added to the standard Tikhonov regularization problem to enforce smoothness of the solution relative to the manifold (in the intrinsic space of the problem) as well as relative to the ambient input space. The minimization problem becomes